

Nouraldin Farge

SOFTWARE ENGINEER — DESKTOP & LOCAL-FIRST SYSTEMS

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SUMMARY

Software engineer who shipped three portable Windows products and a source-free engineering case study using React/TypeScript, Rust/Tauri, Electron, Python, and SQLite. Owns architecture, safety boundaries, verification, and release decisions for local-first systems.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Rust, Python, SQL, HTML/CSS

Desktop & Frontend: React, Tauri, Electron, Windows, WebView2, Vite

Data & Integration: SQLite, Node.js, Express, REST, WebSockets, JSON Schema, Zod

Quality & Release: GitHub Actions, CodeQL, Vitest, pytest, Playwright, Cargo/Clippy, SBOMs, SHA-256, provenance attestations

SELECTED PROJECTS

DrawScope | React · Rust/Tauri · Python · SQLite

[Code](#) · [Release](#) · 2026–Present

- Engineered a portable historical-research workbench around a verified 41,598-row SQLite archive spanning six lottery games, with provenance-aware imports and reproducible data workflows.
- Prevented look-ahead bias with walk-forward evaluation across 250 target draws and a 40% held-out segment; enforced TypeScript, Python, and Rust verification gates.

GameVault | React · Rust/Tauri · SQLite

[Code](#) · [Release](#) · 2026–Present

- Developed a portable game library and launcher with a user-configurable managed root, SQLite persistence, launch history, backups, and optional allowlisted store metadata.
- Designed a review-gated ZIP pipeline that tests archives before extraction, isolates staging, blocks traversal and link/reparse entries, and requires approval before promotion.

Day-Trading Teacher | React · Rust/Tauri · SQLite

[Code](#) · [Release](#) · 2026–Present

- Created a local-first educational environment for trade planning, historical replay, paper trading, journaling, lesson delivery, and evidence-based review.
- Implemented deterministic risk calculations, schema-validated imports, provenance, and controls that exclude live signals, investment advice, and order execution.

Research Studio | Electron · React · TypeScript · Express · SQLite

[Code](#) · 2026–Present

- Architected a guarded AI-assisted enrichment workflow using read-only source access, batches of at most five, schema/evidence validation, human approval, and recoverable exports.

EXPERIENCE

Independent Software Engineer

2026–Present

Portfolio Products

- Shipped three portable Windows releases and one source-free case study; owned requirements, architecture, trust boundaries, automated verification, documentation, and release approval.
- Used AI agents for implementation support while personally reviewing changes, resolving failures, validating claims, and making safety, licensing, data-source, and release decisions.

Technical Assistant (Additional Experience)

Jun 2015–Sep 2015

Chicago Public Schools

Chicago, IL

- Supported staff technology and classroom systems, diagnosed hardware and software issues, and communicated practical resolutions to non-technical users.

EDUCATION

Northeastern Illinois University

May 2023

B.A. Computer Science, Cum Laude — GPA: 3.74

Chicago, IL